Online Event Management System

https://github.com/aiyshwary/Online-Event-Management-System

SQL NoSQL Database Design Normalization

OVERVIEW

A database-driven system for managing events end-to-end, allowing customers to register, browse and book events with discounts, while giving administrators full control over event listings, users, payments and preferences.

IMPACT

Designed and implemented a normalized relational database and NoSQL schema for an Online Event Management System, covering customer registration, booking workflows, discount management, and payment processing.

TECHNICAL WALKTHROUGH

Online Event Management System – Project Explanation This project is a database-driven Online Event Management System designed to help users register, browse, and book events online, while allowing admins to manage events, users, discounts, and payments efficiently.

Problem it Solves
Managing events manually is time-consuming and error-prone—especially

when handling

Multiple event types

User registrations

Discounts and special preferences

Payments and bookings
This system provides a centralized, structured database that automates all these processes.

Key Actors

User

Registers using email and password
Browses events (Dance, Music, Arts, Sports, etc.)

Books events with number of seats

Applies discounts and special preferences

Makes payment using multiple modes

Can update or cancel bookings

Admin

Logs in with admin credentials

Adds, updates, or deletes

Events

Discounts

Special preferences

User details

Manages pricing and discount rates
Core Database Modules (Entities)

User

Stores registered user details

User ID, Email, Username, Password
Users must register before booking any event.

Event Details

Stores complete event information

Event name, type, date & time

Duration and base price
Each event belongs to a specific event category.

Event Discount

Manages discount options

Premium account

Group booking

Coupon-based discounts
Each discount has a rate reduced value.

Special Preference

Optional add-ons for users

VIP seats

Backstage access

Exclusive services
These come with extra charges.

User Booking

Acts as the core linking table

Connects User, Event, Discount, and Special Preference

Stores number of seats booked
Each booking has a unique Booking ID (BID) This table enforces referential integrity using foreign keys.
Amount (Payment)

Handles payment details

Payment ID

Booking ID
Mode of payment (Credit, Debit, Wallet, etc.)

Admin
Stores admin login credentials and permissions.

Database Design Highlights
Fully normalized (3NF) to avoid redundancy

Strong foreign key relationships

Uses indexes for faster search

Supports views for business insights

Important SQL Features Used

Queries

- **i)** Simple queries (filter, sort)
- **ii)** Aggregate queries (COUNT, SUM, AVG)
- **iii)** Join queries (INNER, LEFT, RIGHT joins)

Views

Examples

- **i)** Total amount payable by each user (including discounts & extra
- **ii)** charges)

Users who registered but never booked

Number of bookings per user

Events sorted by price for easy searching

Indexes

Faster event search by type & name

Unique email constraint for users

Faster lookup for bookings, discounts, and prices
Overall Summary (One-Line)
This project implements a relational database–based Online Event Management System that efficiently handles user registrations, event bookings, discounts, special preferences, and payments with strong data integrity and optimized query performance.

KEY DETAILS

1. Designed a fully normalized relational schema (up to 3NF) with entities for Users, Events, Bookings, Payments, and Discounts.
2. Modeled equivalent NoSQL document structures for flexible event-type storage and preference handling.

3. Admin interface supports CRUD operations on events, user management, and discount configuration.
4. Customer portal enables account creation, event browsing, booking with preference selection, and payment tracking.
5. Schema documented with ER diagrams, normalization steps, and a detailed project report.